

Total No. of Questions : 8]

SEAT No. :

P796

[Total No. of Pages : 2

[5870] - 1110
T.E. (Chemical)
CHEMICAL TECHNOLOGY - II
(2019 Pattern) (Semester - I) (309342)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.No.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.
- 4) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables is permitted.
- 5) Figures to the right indicate full marks.

- Q1) a)** Explain the process of manufacturing of Portland cement with major engineering problems. **[9]**
- b) Describe various steps involved in pyrolysis of coal. **[8]**

OR

- Q2) a)** Explain carbonization of coal and the effect of temperature on the products obtained. **[9]**
- b) Explain construction and working of Blast furnace. **[8]**

- Q3) a)** Describe different types of solvents used in paint manufacture? **[9]**
- b) Explain production of water gas and its applications. **[9]**

OR

- Q4) a)** Write a short note on the following with suitable diagram (any 2) **[12]**
- i) Producer Gas
 - ii) Natural Gas
 - iii) Water Gas
- b) Write the classification of dyes with their properties. **[6]**

P.T.O.

Q5) a) Discuss different types of agrochemicals and explain manufacturing process of pesticides. **[9]**

b) Discuss properties and applications of Varnishes. **[8]**

OR

Q6) a) Explain manufacturing of penicillin with major engineering problems. **[9]**

b) Describe in detail about plant growth regulators and yield stimulators. **[8]**

Q7) a) Explain production of methanol in brief with a neat process flow diagram. **[9]**

b) Describe production of styrene with its engineering problems. **[9]**

OR

Q8) Draw a neat flow diagram and explain in brief (any 3) **[18]**

a) Production of vinyl chloride

b) Production of Phenol

c) Production of acetone

d) Production of Ethylene



Total No. of Questions : 8]

SEAT No. :

PA-2652

[Total No. of Pages : 2

[5926]-24
T.E. (Chemical)
CHEMICAL TECHNOLOGY - II
(2019 Pattern) (Semester - I) (309342)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Assume suitable data, if necessary.*
- 4) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables is permitted.*

Q1) a) Explain the process of manufacturing of Portland cement. Write the different types of Portland cement. **[9]**

b) Draw and explain coke oven construction with its applications. **[9]**

OR

Q2) a) Describe carbonization of coal and the effect of temperature on the products obtained? **[9]**

b) Explain construction and working of Blast furnace. **[9]**

Q3) a) Explain production of Producer gas and its applications. **[9]**

b) Explain Mechanical and optical properties of surface coatings. **[8]**

OR

Q4) a) Explain the manufacturing process of dyes with unit operations and unit processes involved in it. **[9]**

b) Discuss production of oxygen gas with its major engineering problems. **[8]**

Q5) a) Explain properties and applications of Varnishes. **[9]**

b) Write short note on agrochemicals and their applications. **[9]**

OR

P.T.O.

- Q6)** a) Describe manufacturing of penicillin with major engineering problems. [9]
b) Enlist Antibiotics with their properties and applications. [9]
- Q7)** a) Explain manufacturing of ethylene dichloride (EDC) with major engineering problems. [9]
b) Explain production of methanol by catalytic hydrogenation of carbon monoxide. [8]

OR

- Q8)** a) Explain production of vinyl chloride production from ethylene dichloride. [9]
b) Explain production of phenol by cumene process. [8]

